Math is Playful

Solve a puzzle by starting small and finding the rule between steps.

Name _____ **Date** _____

Tonight's spark

A game-night sign lists the fewest moves for a tower. One disc: 1. Two discs: 3. Three discs: 7. Someone erased the row for four discs. What did it say?

Guess first: more than 10 moves, or fewer?

Try it

Finish each sentence in your own words.

1. **My rule goes from ___ to ___ by ___ and ___.**

   My words _____

2. **In my table, the first column is ___ and the second is ___.**

   My words _____

3. **Using the rule, I predict the next number is ___.**

   My words _____

4. **Each number in the pattern is ___ the one before.**

   My words _____

Talk about it

Try your rule on two discs. Does 1 doubled give the real number of moves?

Home mission

Count out dry beans: a pile of 1, a pile of 3, and a pile of 7. Each pile is double the last plus one. Before counting, ask your student how many beans the next pile needs. Then count to test.

If we get stuck, ask

- What is the smallest version of this problem? Solve that first.
- Can you put your results in a table?
- What happens between one row and the next?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/10-3-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it